

Lecture 21

Linear Transformation II

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Outline and Review

5 mins

Roadmap of Semester

Segment 1 Solving linear systems (Lec 3-15);

Segment 2 **Lec 16-22:** Three relatively independent parts:

- **Orthogonality**. Least squares; orthogonal matrix (Lect 16,17,18)
- **Determinant**. [Important tool!] Lec 18,19, 20
- **Linear transformation**. Lec 20-21 [Advanced math perspective of matrix]

They are not directly related to solving system/problem,
but are **fundamental and useful tools**.

Segment 3 **Lec 22-28: Eigenvalues and singular values.**

Today's Lecture: Outline

Main topic: Linear transformation II

Strang's book: Sec 8.1, 8.2

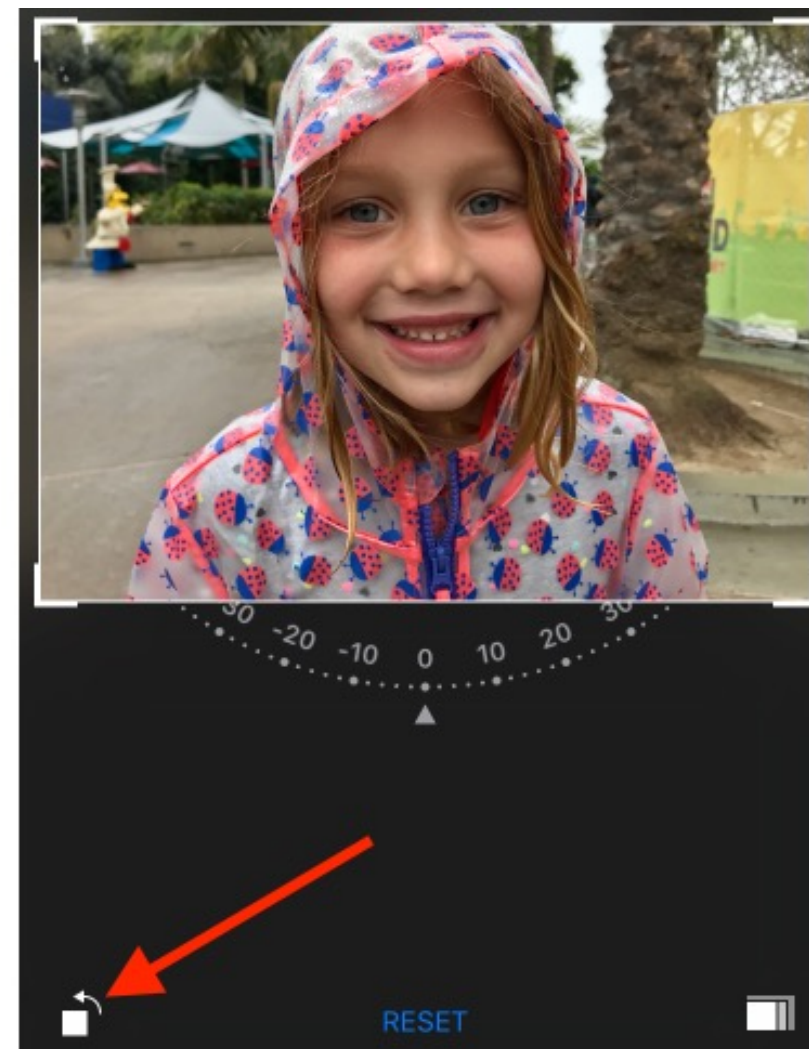
1. View Matrix as Linear Transformation
2. Linear Transformation on General Linear Spaces
3. Matrix Representation of General Linear Transformation

After the lecture, you should be able to

1. Tell the relation of linear transformation and matrix
2. Verify linear transformation and compute it for **general** linear spaces
3. Derive the matrix representation of general linear transformation

Rotate Photos

How to rotate photos on iPhone



Motivating Question

Yes, I know you can do it on the phones.

How do the phones accomplish the job?



Question 1: Is it a **linear** transformation?

Question 2: How do I **derive** its expression?

Linear Transformation

Definition 20.1 (linear transformation):

If a mapping f from a linear space V to a linear sapce W satisfies

$$f(\alpha \mathbf{x} + \beta \mathbf{y}) = \alpha f(\mathbf{x}) + \beta f(\mathbf{y}), \forall \alpha, \beta \in \mathbb{R}, \forall \mathbf{x}, \mathbf{y} \in V,$$

Then we say f is a **linear transformation** from V to W .

In words, **transformed LC of vectors** = **LC of transformed vectors**.

Example:

Suppose $A \in \mathbb{R}^{m \times n}$ is a given real matrix.

$f(\mathbf{x}) = A\mathbf{x}$ is a **linear transformation** from $\mathbb{R}^n$ to $\mathbb{R}^m$.

Theorem 20.1

For any linear mapping f from $\mathbb{R}^n$ to $\mathbb{R}^m$, there exists a matrix A such that $f(x) = Ax$.

Verify Linear Transformation by Geometry

$$\Omega_1 \triangleq \{\alpha \mathbf{x} + \beta \mathbf{y} \mid \alpha + \beta = 1\} \quad \Omega_2 \triangleq \{\alpha f(\mathbf{x}) + \beta f(\mathbf{y}) \mid \alpha + \beta = 1\}$$

By definition of linear transformation: $f(\Omega_1) = \Omega_2$, i. e.,

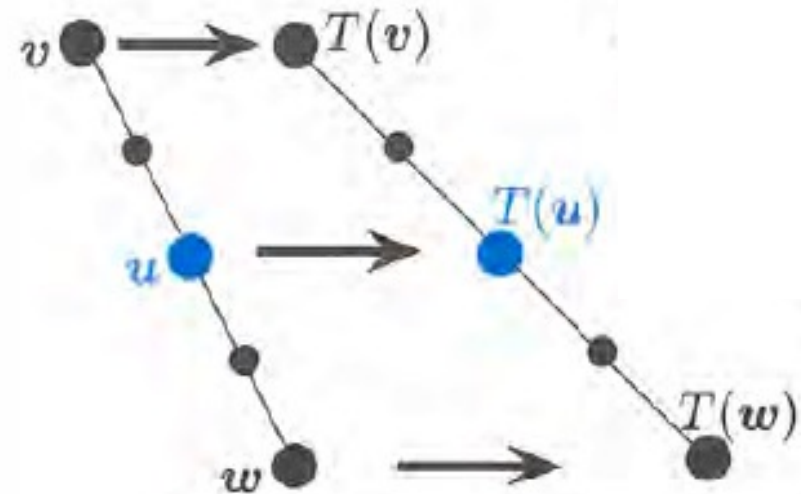
$$\alpha + \beta = 1$$

Rule 1: Lines to Lines.

Rule 2: Equally spaced points to equally spaced points.

$$\alpha = \beta = 0$$

Rule 3: Origin to Origin.



$$\Omega_1 \triangleq \{\alpha \mathbf{w} + \beta \mathbf{v} \mid \alpha + \beta = 1\} \quad \Omega_2 \triangleq \{\alpha T(\mathbf{w}) + \beta T(\mathbf{v}) \mid \alpha + \beta = 1\}$$

Euclidean Space: Computing Matrix of Linear Transformation

Algorithm 20.1

Step 1: Pick the standard basis $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ of $\mathbb{R}^n$.

Step 2: Compute $f(\mathbf{e}_i) \in \mathbb{R}^m, \forall i$.

Step 3: Form a matrix $A = [f(\mathbf{e}_1), f(\mathbf{e}_2), \dots, f(\mathbf{e}_n)] \in \mathbb{R}^{m \times n}$

Conclusion: For any $\mathbf{x} \in \mathbb{R}^n$, we have

$$f(\mathbf{x}) = \sum_i x_i f(\mathbf{e}_i) = A\mathbf{x}.$$

A is the matrix of the linear transformation f .

Claim: Suppose f is a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$.

Let $A = [f(\mathbf{e}_1), f(\mathbf{e}_2), \dots, f(\mathbf{e}_n)]$, then $f(\mathbf{x}) = A\mathbf{x}$.

Input: linear transformation f .

Output: matrix A , whose columns are $f(\mathbf{e}_i)$'s.

Part I View Matrix as Linear Transformation

15 mins

Transformation $\rightarrow$ Matrix

Recall:

Claim: Suppose f is a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$.

Let $A = [f(\mathbf{e}_1), f(\mathbf{e}_2), \dots, f(\mathbf{e}_n)]$, then $f(\mathbf{x}) = A\mathbf{x}$.

Start: linear transformation f .

End: matrix A ,

Relation: A 's columns are $f(e_i)$'s.

$$f: \mathbb{R}^n \rightarrow \mathbb{R}^m \longrightarrow A = [f(\mathbf{e}_1), \dots, f(\mathbf{e}_n)] \in \mathbb{R}^{m \times n}$$

Transformation $\rightarrow$ Matrix

Recall:

Claim: Suppose f is a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$.

Let $A = [f(\mathbf{e}_1), f(\mathbf{e}_2), \dots, f(\mathbf{e}_n)]$, then $f(\mathbf{x}) = A\mathbf{x}$.

Start: linear transformation f .

End: matrix A ,

Relation: A 's columns are $f(\mathbf{e}_i)$'s.

$$f: \mathbb{R}^n \rightarrow \mathbb{R}^m \longrightarrow A = [f(\mathbf{e}_1), \dots, f(\mathbf{e}_n)] \in \mathbb{R}^{m \times n},$$

Reverse perspective.

Suppose A is an $m \times n$ matrix.

Start: matrix A .

Then $f(x) = Ax$ is a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$,

End: transformation f .

and $[f(\mathbf{e}_1), \dots, f(\mathbf{e}_n)] = A$, i.e., $f(\mathbf{e}_i)$ is the i -th column of A .

Relation: $f(\mathbf{e}_i)$'s are A 's column

$$A = [\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n] \longrightarrow f: \mathbb{R}^n \rightarrow \mathbb{R}^m \text{ where } f(\mathbf{e}_i) = \mathbf{a}_i$$

Transformation $\rightarrow$ Matrix

Matrix $A = \begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix} = [\vec{a}_1 \quad \vec{a}_2]$ $A \vec{x}$.

Multiplying by $A : \vec{e}_1, \vec{e}_2 \xrightarrow{A} \text{columns of } A$.

$$A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad (\text{Column 1})$$

$$A \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix} \quad (\text{Column 2})$$

$$\begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Any vector $\vec{x}$:

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_1 \vec{e}_1 + x_2 \vec{e}_2 \xrightarrow{A} A\vec{x} = x_1 \vec{a}_1 + x_2 \vec{a}_2$$

Conclusion: Matrix A defines a linear transformation $A\vec{x}$.

It maps $\vec{e}_1, \vec{e}_2$ to $\vec{a}_1, \vec{a}_2$ (columns of A).

Important New View of Matrix

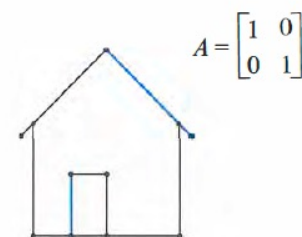
Previous view: Matrix can represent data. (e.g. image)

Recall: $f(\mathbf{x}) = A\mathbf{x}$ is a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$

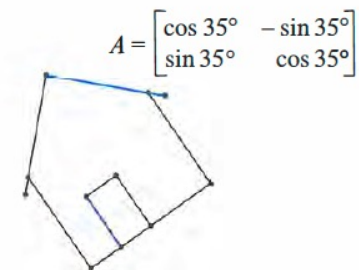
New View: Matrix can represent linear transformation.

To better understand it,
we will “see” the transformation defined by a matrix.

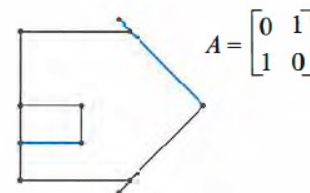
Remark: Matrix is a math tool.
Matrix can represent data;
Matrix can represent transformation.



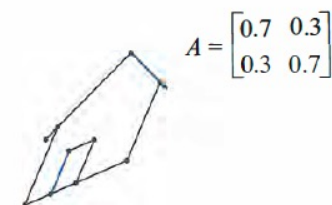
$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$



$$A = \begin{bmatrix} \cos 35^\circ & -\sin 35^\circ \\ \sin 35^\circ & \cos 35^\circ \end{bmatrix}$$



$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$



$$A = \begin{bmatrix} 0.7 & 0.3 \\ 0.3 & 0.7 \end{bmatrix}$$

Transformation feature 1: Basis Changes

Matrix can represent a linear transformation.

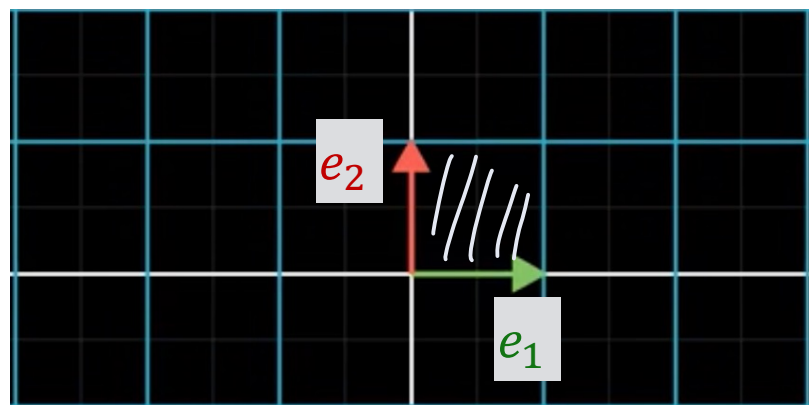
We visualize the transformation defined by a matrix.

e.g. Matrix A:

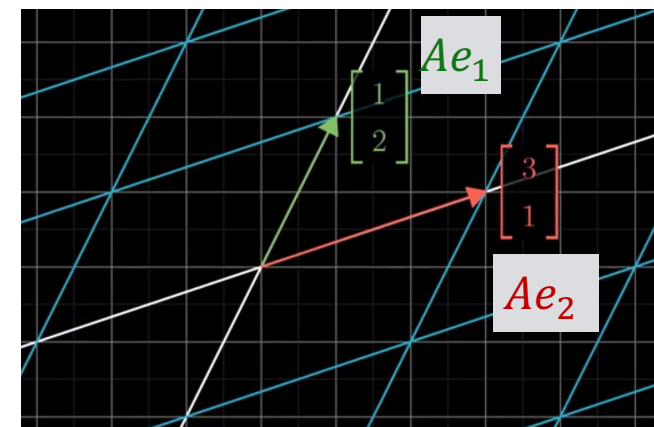


Columns of the matrix $\rightarrow$ transformed basis vectors.

Linear transformation Ax :



Input space:
"Square tile"



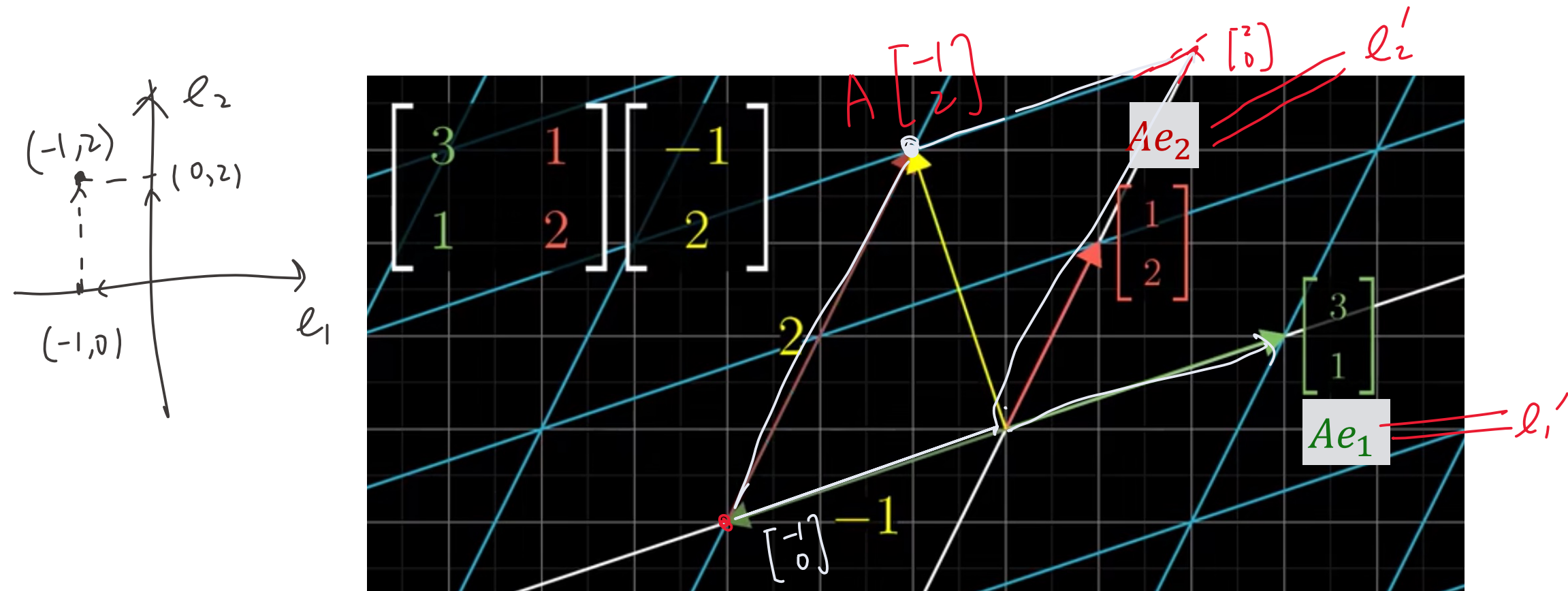
Output space:
"Parallelogram tile"

Transformation feature 2: coordinate stay same

$$A: (-1, 2) \mapsto (-1) Ae_1 + 2 Ae_2$$

Geometrically:

A maps $(-1, 2)$ to a point **with coordinate $(-1, 2)$** in the new system.



The matrix A changes the coordinate system:

it stretches and rotates the basis vectors, and then re-expresses $(-1, 2)$ in that new basis.

Matrix $\rightarrow$ Transformation: Dependent Columns

Dependent columns of the matrix

$\rightarrow$ transformed “basis vectors” are degenerate

Matrix A:

$$\begin{bmatrix} 2 & -2 \\ 1 & -1 \end{bmatrix}$$

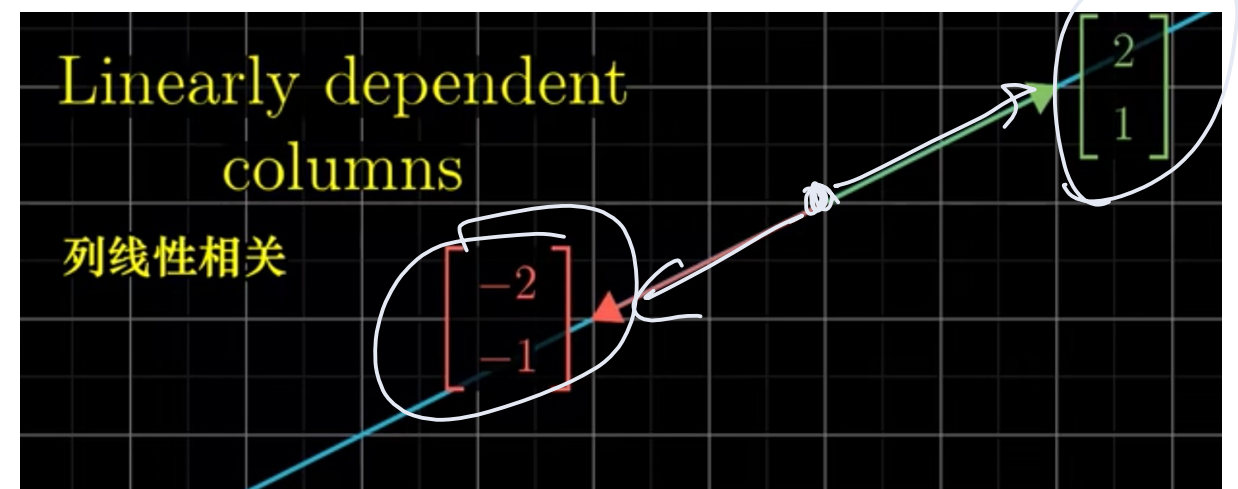
$$A: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$
$$x \mapsto Ax$$

The set $\{Ax: x \in \mathbb{R}^2\}$

= Column space of A

$$= \{\alpha \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \beta \begin{bmatrix} -2 \\ -1 \end{bmatrix} : \alpha, \beta \in \mathbb{R}\} = \text{span}\left(\begin{bmatrix} 2 \\ 1 \end{bmatrix}\right)$$

Linear transformation Ax :

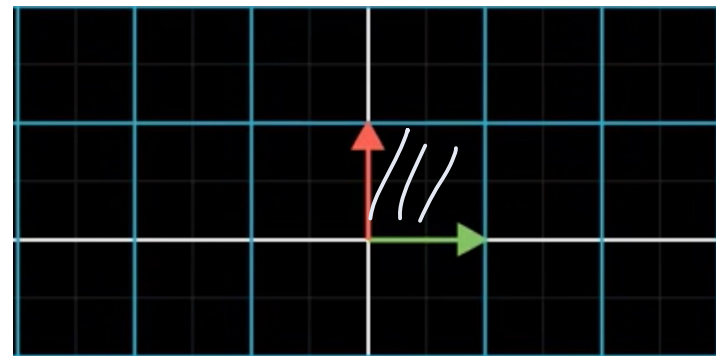


Determinant: Geometry Interpretation II

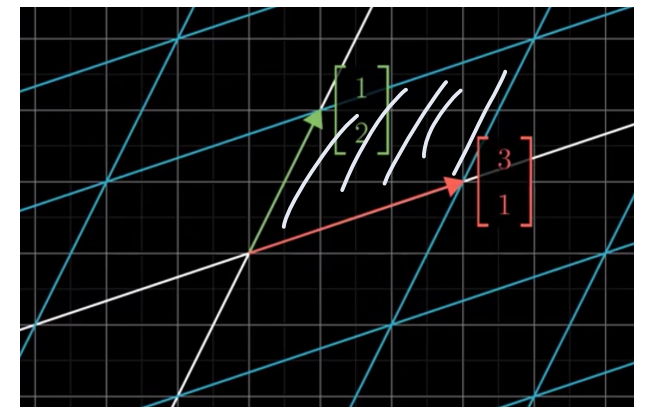
Matrix A:



Linear transformation Ax :

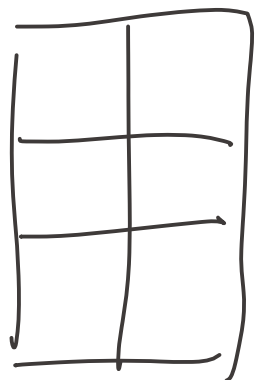


Input space: *Area 1*
"Square tile"



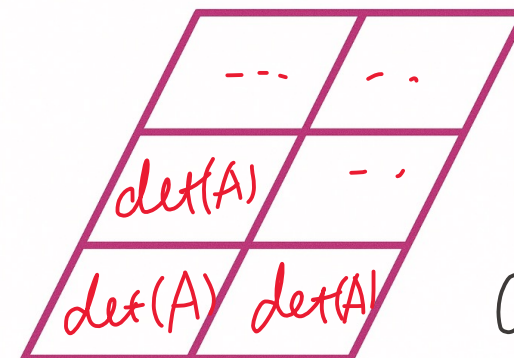
Output space:
"Parallelogram tile"
area $\det(A)$

Lec 19: Area of $\text{PARA}([a_1, a_2])$ is $\det(A)$.



Area 6

Volume changes
by $\det(A)$ times

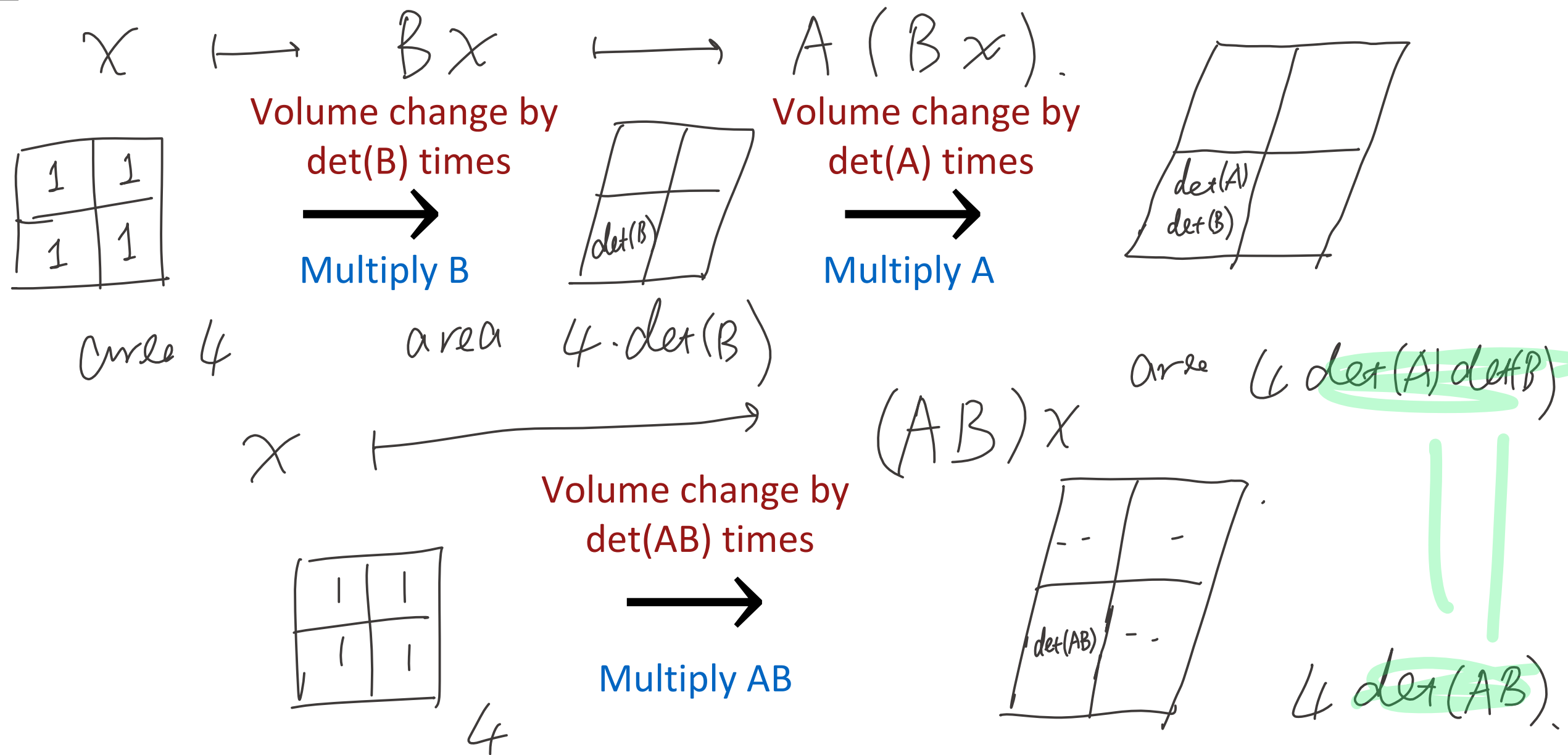


Area $6 \cdot \det(A)$

Linear transformation perspective:

$\det(A)$ = area of $\text{PARA}([a_1, a_2])$ = volume change ratio of linear transformation A

Determinant: Geometry Interpretation II



Geometrical interpretation of $\det(AB) = \det(A) \det(B)$

Part II Verify Linear Transformation between Non-Euclidean Spaces

40 mins

Linear Spaces

Definition 20.1 (linear transformation):

If a mapping f from a linear space V to a linear sapce W satisfies

$$f(\alpha \mathbf{x} + \beta \mathbf{y}) = \alpha f(\mathbf{x}) + \beta f(\mathbf{y}), \forall \alpha, \beta \in \mathbb{R}, \forall \mathbf{x}, \mathbf{y} \in V,$$

Then we say f is a **linear transformation** from V to W .

All examples we've seen so far:
 V and W are Euclidean spaces.

Next:

Verify linear transformation when V or W are NOT
Euclidean spaces.

Part II.1 Matrix Space

10 mins

Simple Examples on Matrix Space

Eg: Suppose $X \in \mathbb{R}^{m \times n}$ is a matrix variable.

Then $f(X) = 2X$ is a linear transformation from $\mathbb{R}^{m \times n}$ to $\mathbb{R}^{m \times n}$.

$$\begin{array}{cc} f(\alpha X) \stackrel{?}{=} \alpha f(X) & f(X+Y) \stackrel{?}{=} f(X) + f(Y) \\ \parallel & \parallel \end{array}$$

$$2(\alpha X) \stackrel{=}{=} \alpha \cdot (2X)$$

since scalar α can be anywhere

$$2(X+Y) \stackrel{=}{=} 2X + 2Y$$

distributive rule of scalar times matrix

Eg: Suppose $A \in \mathbb{R}^{k \times m}$ is fixed. $X \in \mathbb{R}^{m \times n}$ is a matrix variable.

Then $f(X) = AX$ is a linear transformation from $\mathbb{R}^{m \times n}$ to $\mathbb{R}^{k \times n}$.

$$\begin{array}{cc} f(\alpha X) \stackrel{?}{=} \alpha f(X) & f(X+Y) \stackrel{?}{=} f(X) + f(Y) \\ \parallel & \parallel \end{array}$$

$$A(\alpha X) \stackrel{=}{=} \alpha (AX)$$

scalar can be anywhere

$$A(X+Y) \stackrel{=}{=} AX + AY$$

distributive rule of matrix multiplication

Examples or Non-Examples on Matrix Space

Exercise: Is “inverse” a linear transformation?

Suppose $X \in \mathbb{R}^{n \times n}$ is a matrix variable.

Then $f(X) = X^{-1}$ is NOT a linear transformation.

Method 1 $(\alpha X)^{-1} = \frac{1}{\alpha} X^{-1} \neq \alpha X^{-1}$ (if $\alpha \neq 0, \alpha \neq 1$).

Method 2 f is NOT defined for some X on $\mathbb{R}^{n \times n}$,

so f is NOT a mapping
from $\mathbb{R}^{n \times n} \rightarrow \mathbb{R}^{n \times n}$

Exercise: Is “transpose” a linear transformation?

Suppose $X \in \mathbb{R}^{m \times n}$ is a matrix variable.

Then $f(X) = X^T$ is a linear transformation.

$f: \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{n \times m}, \quad X \mapsto X^T.$

$$(\alpha X)^T = \alpha X^T, \quad (X + Y)^T = X^T + Y^T.$$

Part II.2 Example of Polynomial Space

15 mins

Recall: Polynomial Space

Lecture 11 (Linear Space)

Eg: The set of polynomials with degree no more than k is a linear space.

P_k

Remark:

Operations: Addition: $f + g = \sum_{i=0}^k (a_i + b_i)x^i$; Scalar multiplication: $\alpha f = \sum_{i=0}^k (\alpha a_i)x^i$

Lecture 13 (Basis) Part III.2

Find a basis of the vector space P_2 , where P_2 is defined as

$$P_2 \triangleq \{f(x) = a_0 + a_1x + a_2x^2 : a_0, a_1, a_2 \in \mathbb{R}\}.$$

Define the polynomials $f_0(x) = 1$, $f_1(x) = x$, and $f_2(x) = x^2$.

Then the set $\{f_0, f_1, f_2\}$ is a basis of P_2 .

Explanation of Notation: f v.s. $f(x)$

$P_2 = \{\text{polynomials of degree no more than 2}\}$

Shall we write $f \in P_2$, or $f(x) \in P_2$?

-- f denotes a **function** as a whole (e.g., a polynomial).

-- $f(x)$ denotes the **value** of the function of f at a specific input x .
or a general function $f(\cdot)$

--**Strictly speaking**, we should write $f \in P_2$,
because P_2 is a **set of functions**, not a set of values.

--**Formally**: Consider a function $f \in P_2$ where f is defined as $f(x) = x^2 + 1$.

$$\text{or } f(t) = t^2 + 1$$

$$\text{or } f(z) = z^2 + 1.$$

But a bit too long...

--**Short and informal version**:

Consider a function $f(x) = x^2 + 1 \in P_2$.

or consider a function $x^2 + 1 \in P_2$ *or* $x^2 + 1 \in P_2$.

or consider a function $f(t) = t^2 + 1 \in P_2$.

Examples on Polynomial Space

Example Let $T : P_3 \rightarrow \mathbb{R}^{2 \times 2}$ be a mapping defined as:

$$T(\underline{ax^3 + bx^2 + cx + d}) = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Is this T a linear transformation?

Recall Definition: f is a linear transformation from V to W iff

$$f(\alpha \mathbf{x} + \beta \mathbf{y}) = \alpha f(\mathbf{x}) + \beta f(\mathbf{y}), \forall \alpha, \beta \in \mathbb{R}, \forall \mathbf{x}, \mathbf{y} \in V,$$

Incorrect method: Pick a polynomial f , check $f(x + y) = f(x) + f(y)$, $\forall x, y \in P_3$.

Correct: Rewrite definition in this setting carefully. *how to write function x ?*

T is a linear transformation from V to W iff

$$T(\underline{f + g}) = T(\underline{f}) + T(\underline{g}), \quad T(\underline{\alpha f}) = \alpha T(\underline{f}), \quad \forall \underline{f}, \underline{g} \in P_3, \forall \alpha \in \mathbb{R}$$

Think: What **symbols** do we use to represent elements of P_3 ?

Examples on Polynomial Space

[Suggested break]

Example Let $T : P_3 \rightarrow \mathbb{R}^{2 \times 2}$ be a mapping defined as:

$$T(ax^3 + bx^2 + cx + d) = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Solution: We need to show

$$T(f + g) = T(f) + T(g), \quad T(\alpha f) = \alpha T(f), \quad \forall f, g \in P_3, \forall \alpha \in \mathbb{R}$$

Need to represent elements of P_3 .

For $f(x) = \sum_{i=0}^3 a_i x^i$, $g(x) = \sum_{i=0}^3 b_i x^i \in P_3$ and $\alpha \in \mathbb{R}$,

Sum:

$$(f + g)(x) = \sum_{i=0}^3 (a_i + b_i) x^i.$$
$$T(f + g) = \begin{bmatrix} c_3 + b_3 & a_2 + b_2 \\ a_1 + b_1 & a_0 + b_0 \end{bmatrix} = \begin{bmatrix} a_3 & a_2 \\ a_1 & a_0 \end{bmatrix} + \begin{bmatrix} b_3 & b_2 \\ b_1 & b_0 \end{bmatrix} = T(f) + T(g).$$

Scalar multiplication:

$$(\alpha f)(x) = \sum_{i=0}^3 (\alpha a_i) x^i.$$
$$T(\alpha f) = \begin{bmatrix} \alpha a_3 & \alpha a_2 \\ \alpha a_1 & \alpha a_0 \end{bmatrix} = \alpha \begin{bmatrix} a_3 & a_2 \\ a_1 & a_0 \end{bmatrix} = \alpha T(f).$$

Part II.3 Function Space and Differentiation

15 mins

Function Space $C(\mathbb{R})$ and $C^1(\mathbb{R})$

Continuous function space $C(\mathbb{R})$:

The set of continuous functions form a linear space.

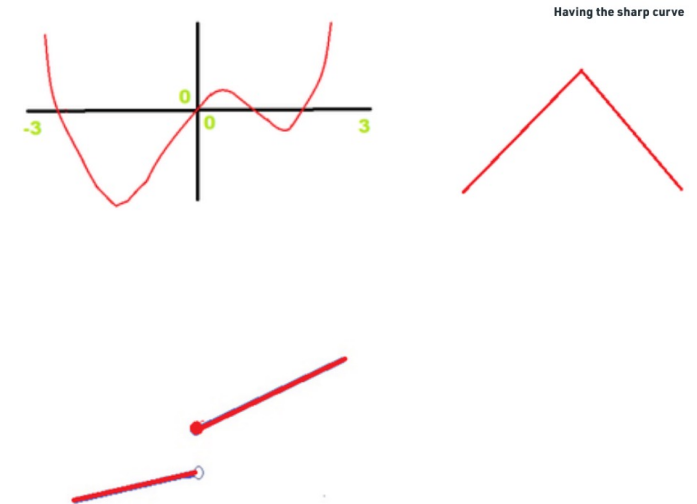
When equipped with the following operations, $C(\mathbb{R})$ forms a **real vector space**:

1. **Function Addition** (vector addition):

$$(f + g)(x) \triangleq f(x) + g(x), \quad \forall f, g \in C(\mathbb{R}), x \in \mathbb{R}.$$

2. **Scalar Multiplication**:

$$(\alpha f)(x) \triangleq \alpha \cdot f(x), \quad \forall \alpha \in \mathbb{R}, f \in C(\mathbb{R}), x \in \mathbb{R}.$$



Function Space $C(\mathbb{R})$ and $C^1(\mathbb{R})$

Continuous function space $C(\mathbb{R})$:

The set of continuous functions form a linear space.

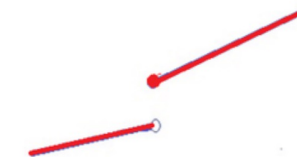
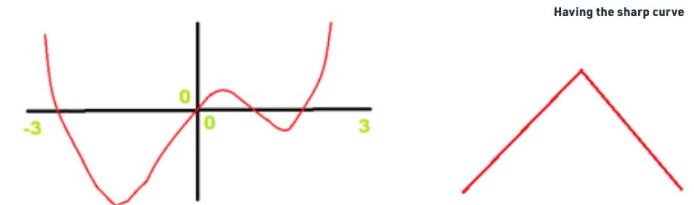
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$$(\alpha f)(x) \triangleq \alpha \cdot f(x), \quad \forall \alpha \in \mathbb{R}, f \in C(\mathbb{R}), x \in \mathbb{R}.$$



Not continuous

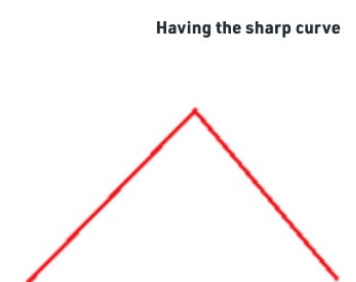
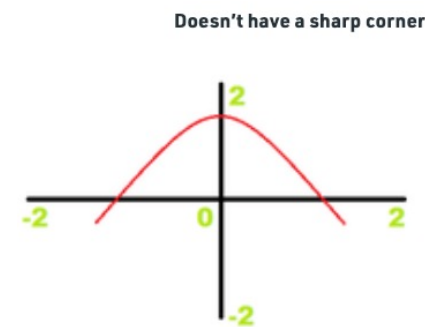
Continuously differentiable function space $C^1(\mathbb{R})$: a linear space.

$$C^1(\mathbb{R}) = \{f : \mathbb{R} \rightarrow \mathbb{R} \mid f'(x) \text{ exists for all } x \in \mathbb{R} \text{ and } f' \text{ is continuous on } \mathbb{R}\}.$$

f has a smooth graph, meaning it does not have any sharp corners.

Need to verify the axioms.

Skip details here.



Not differentiable

Function Space $C(\mathbb{R})$: Explanation of Notation

Continuous function space $C(\mathbb{R})$:

The set of continuous functions form a linear space.

When equipped with the following operations, $C(\mathbb{R})$ forms a **real vector space**:

1. **Function Addition** (vector addition):

$$(f + g)(x) \triangleq f(x) + g(x), \quad \forall f, g \in C(\mathbb{R}), x \in \mathbb{R}.$$

2. **Scalar Multiplication**:

$$(\alpha f)(x) \triangleq \alpha \cdot f(x), \quad \forall \alpha \in \mathbb{R}, f \in C(\mathbb{R}), x \in \mathbb{R}.$$

Here, $(f + g)$ should be a new function h in $C(\mathbb{R})$.

$$(f + g)(x) \triangleq f(x) + g(x)$$

means: the new function $h = f + g$ is defined as $h(x) \triangleq f(x) + g(x)$,
 $\forall x \in \mathbb{R}$.

e.g. $f(x) = x^2 - 1$, $g(x) = -x^2 + x$,

then $\underbrace{(f + g)}_{\text{a function}}(x) = x - 1$.

Is Differentiation a Linear Transformation?

Eg: Is $T(u) = u'$ is a linear transformation? Here u' is the derivative of u .

Reminder of notation: e.g. If $u(x) = x^2$, then $u'(x) = 2x$.

Check: $T(\alpha u) \stackrel{?}{=} \alpha T(u)$ $T(u+v) \stackrel{?}{=} T(u) + T(v)$

T is not defined for all functions.

Before checking linearity,
need to specify a linear space s.t. T is defined!

Definition 20.1 (Linear Transformation)

Let V and W be two linear spaces.

A mapping $L : V \rightarrow W$ is called a linear transformation if

$T(u) = u'$ is
defined for any
element of $C^1(\mathbb{R})$.

Differentiation on $C^1(\mathbb{R})$ is Linear Transformation

Claim: $T(u) = u'$ defined on $C^1(\mathbb{R})$ is a mapping from $C^1(\mathbb{R})$ to $C(\mathbb{R})$.
It is a linear transformation.

Verify: Well-defined: If $f \in C^1(\mathbb{R})$, then $T(f) = f' \in C(\mathbb{R})$ by definition of C^1 .

Thus T is a mapping from a linear space $C^1(\mathbb{R})$ to $C(\mathbb{R})$.

$$T : C^1(\mathbb{R}) \rightarrow C(\mathbb{R}), \quad f \mapsto f'$$

For any $f, g \in C^1(\mathbb{R})$ and scalar α ,

$$T(f + g) = (f + g)' = f' + g' = T(f) + T(g)$$

$$T(\alpha f) = (\alpha f)' = \alpha f' = \alpha T(f) \quad \square$$

e.g. For $f(x) = x^2$ and $g(x) = \sin(x)$,

In calculus class, you compute $(x^2 + \sin x)' = 2x + \cos x$.

In algebra class, we remind you: you are using linearity of differentiation!

$$T(x^2 + \sin x) = T(f + g) = T(f) + T(g) = T(x^2) + T(\sin x) = 2x + \cos x.$$

Differentiation on P_2 is Linear Transformation

Claim: $T(u) = u'$ defined on P_2 is a mapping from P_2 to P_1 .

It is a linear transformation.

$$(x^2)' = 2x, \quad (x^2 - x)' = 2x - 1.$$

The proof is similar to the problem for u' defined on $C^1(\mathbb{R})$.

Skip here.

There is a closed form expression of the transformation, since it is only defined on P_2 :

$$T(ax^2 + bx + c) = (ax^2 + bx + c)' = 2ax + b.$$

Part III Matrix Representation of Linear Transformation for Any Linear Spaces

10 mins

Relation of Matrix and Linear Mapping in General Space?

Euclidean Space

For any linear mapping

$$f: \mathbb{R}^n \rightarrow \mathbb{R}^m$$

$\exists$ a matrix A such
that $f(x) = Ax$.

General Space

For any linear mapping

$$f: V \rightarrow W,$$



Matrix Representation of Differentiation

$T(u) = u'$ defined on P_2 is a linear transformation from P_2 to P_1 .
Is T related to a matrix?

Try to apply Algorithm 20.1?

Algorithm 20.1

Step 1: Pick a basis $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ of $\mathbb{R}^n$. $\{1, x, x^2\}$
 $\{v_1, v_2, \dots, v_n\}$ of P_2 .

Step 2: For all i , compute $f(\mathbf{e}_i) \in \mathbb{R}^m$ $T(v_i) \in W$ $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $x', (x^2)'$ $\begin{bmatrix} 1 \\ 2x \end{bmatrix}$

Step 3: Form $A = [T(v_1), T(v_2), \dots, T(v_n)] \in \mathbb{R}^{m \times n}$

Conclusion: For any $\mathbf{x} \in \mathbb{R}^n$ $\mathbf{x} \in P_2$, we have
 $\dots \dots T\mathbf{x} = A\mathbf{x}$.
 $[0, 1, 2x]$
 $\downarrow$ function

Vote: Which step is not valid?

A. Step 1

B. Step 2

C. Step 3

D. All are valid

Matrix Representation of Differentiation

$T(u) = u'$ defined on P_2 is a linear transformation from P_2 to P_1 .
Is T related to a matrix?

Try to apply Algorithm 20.1?

Algorithm 20.1

Step 1: Pick a basis $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ of $\mathbb{R}^n$. $\{v_1, v_2, \dots, v_n\}$ of P_2 .

Step 2: For all i , compute $f(\mathbf{e}_i) \in \mathbb{R}^m$ $T(v_i) \in W$

Step 3: Form $A = [T(v_1), T(v_2), \dots, T(v_n)] \in \mathbb{R}^{m \times n}$

Conclusion: For any $\mathbf{x} \in \mathbb{R}^n$ $\mathbf{x} \in P_2$, we have
... ..

Challenge: the collection of $T(v_i)$ is NOT a matrix, if $T(v_i)$ are not vectors.
e.g. polynomials
[$x - 1$, $x^2 + x - 2$] is NOT a matrix.

Solution:

Represent $T(v_i)$ as a vector.

Recall: Vector Representation of Linear Space

Proposition (Unique Representation via Basis)

Let $\mathcal{U} = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be a basis of a linear space X . [presented before]

Any $x \in X$ can be **uniquely** represented as a linear combination of $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$.

e.g. Consider P_2 .

A basis of P_2 :

$$\mathcal{B} = \{1, x, x^2\}.$$

Representing a
polynomial in P_2
by this basis:

$$f = a_0 + a_1x + a_2x^2 = \begin{bmatrix} 1 & x & x^2 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix}$$

Simple expression of f
by a vector:

$$[f]_{\mathcal{B}} = \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix}$$

General Space: Computing Matrix of Linear Transformation

Consider a linear transformation $T: V \rightarrow W$.
Find a matrix representation of T .

Algorithm 21.1

Step 1: Pick a basis $Y = \{v_1, \dots, v_n\}$ of the domain V ;
Pick a basis $Z = \{w_1, \dots, w_m\}$ of the codomain W .

Step 2: Express $T(v_j) = a_{1j}w_1 + \dots + a_{mj}w_m \in W, \forall j$.
Write the coordinates of $T(v_j)$ as $a_j = (a_{1j}, \dots, a_{mj})^T$

Step 3: Form a matrix $A = [a_1, \dots, a_n] \in \mathbb{R}^{m \times n}$

Conclusion: A is a matrix representation of T
with respect to the basis Y of V and the basis Z of W .

Remark: For simplicity, can also say:

A is a matrix representation of T w.r.t. (the bases) Y and Z .

Matrix Representation of Differentiation

We know $T(u) = u'$ defined on P_2 is a linear transformation from P_2 to P_1 . Find a matrix representation of T .

Solution Method 1:

Step 1: Identify a basis of input space P_2 : $Y = \{1, x, x^2\}$;
a basis of output space P_1 : $Z = \{1, x\}$;

Step 2: Compute the coordinates of $T(\text{basis vectors})$ under the basis Z .

$$\begin{array}{ll} T(1) = 1' = 0 = 0 \cdot 1 + 0 \cdot x, & \text{Coordinates } [0, 0]^T. \\ T(x) = x' = 1 = 1 \cdot 1 + 0 \cdot x, & \text{Coordinates } [1, 0]^T. \\ T(x^2) = (x^2)' = 2x = 0 \cdot 1 + 2 \cdot x. & \text{Coordinates } [0, 2]^T. \end{array}$$

Step 3: Form a matrix by combining the coordinate vectors in Step 2.

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Conclusion: The matrix representation of T w.r.t. Y and Z is

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Interpretation of the Matrix Representation

We get the matrix of $T(u) = u' : P_2 \rightarrow P_1$

So what?

How does this matrix help us understand the transformation?

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} = \text{matrix form of the derivative } T : u \mapsto u'.$$

Input $u = \underline{a} + \underline{b}x + \underline{c}x^2$ $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$ $Au = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} b \\ 2c \end{bmatrix}$ Output $u' = b + 2cx$ $\begin{bmatrix} b \\ 2c \end{bmatrix}$

Once we get coordinates in both spaces,
we can view the transformation T as a mapping from “**vectors to vectors**”;
and the **matrix A describes this mapping**.

Diagram: View L.T. as Operating on Coordinates

$$V \longrightarrow W$$

$$x = \sum_{i=1}^n x_i \mathbf{v}_i \mapsto \mathbf{y} = T(x) = \sum_{j=1}^m y_j \mathbf{w}_j$$

Coordinates of x

Coordinates of $y = T(x)$

$$(x_1, \dots, x_n) \xrightarrow{\text{Multiply } A} (y_1, \dots, y_n)$$

Can infer A from this relation

e.g. $\begin{bmatrix} a \\ b \\ c \end{bmatrix} \xrightarrow{\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}} \begin{bmatrix} b \\ 2c \end{bmatrix}$

Once we get coordinates in both spaces,
We can view the linear transformation
as a mapping from “vectors to vectors”



Summary Today (Write Your Own)

One sentence summary:

Detailed summary:

Summary Today (Write Your Own)

One sentence summary:

We learned general linear transformation and how to “express” it.

Detailed summary:

Matrix can be viewed as linear transformation.

- Transform basic building block of input space to a polytope build block.
- $\det(A)$ = volume change ratio of the transformation

Linear transformation between two linear spaces.

- Matrix space: transpose is a linear transformation.
- Function space: Differentiation is a linear transformation.

—Matrix representation of general linear transformation:

Method 1: Step 1: Find the transformed input basis elements;

Step 2: Find their vector representations under output basis;

Step 3: Combine these vectors to get a matrix. .